



Describing a translation

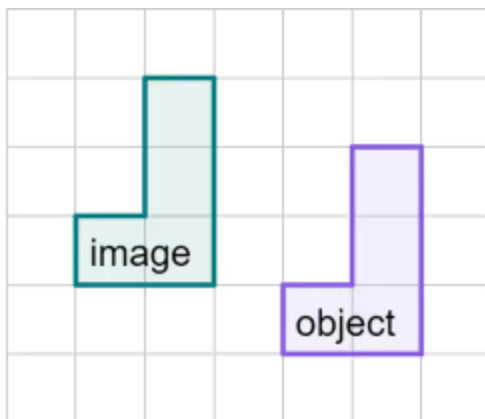
1 Displacement is the distance from the starting point when measured in a straight line. Fill in the blank

2 $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ is a column vector. It has two numbers. Match each number with its meaning. Write the correct letter in each box

a	The top number, 2, is the
b	The bottom number, 1, is the

b	vertical displacement
a	horizontal displacement

3 Which vector matches this translation? Tick 1 correct answer



☒ $\begin{pmatrix} -3 \\ 1 \end{pmatrix}$

☐ $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$

☐ $\begin{pmatrix} 3 \\ -1 \end{pmatrix}$

☐ $\begin{pmatrix} -2 \\ 1 \end{pmatrix}$

☐ $\begin{pmatrix} 1 \\ -3 \end{pmatrix}$



- 4 Which of the following shows a translation by $\begin{pmatrix} 2 \\ -3 \end{pmatrix}$? Tick **1** correct answer



- 5 Match the vector to the description. Write the correct letter in each box

a	$\begin{pmatrix} -4 \\ 1 \end{pmatrix}$
b	$\begin{pmatrix} 4 \\ 1 \end{pmatrix}$
c	$\begin{pmatrix} -4 \\ -1 \end{pmatrix}$
d	$\begin{pmatrix} 4 \\ -1 \end{pmatrix}$

b	right by 4 units and up by 1 unit
c	left by 4 units and down by 1 unit
a	left by 4 units and up by 1 unit
d	right by 4 units and down by 1 unit

- 6 A triangle ABC is translated. The corresponding coordinates are A(1,5) and A'(2,7), B(5,8) and B'(6, 10). C has coordinates (4,7). Work out the coordinates of the image of C. Fill in the blank

(5, 9)

